

COURSE DESCRIPTION

Course Number	0418 - 201	Course Title	Data Structure and Algorithm
Credit Hours	3 (3:0:3)	Course Coordinator	Dr. Bader AlBader
Compulsory	Yes	Prerequisite	0418-142

Textbook

Title:	Data Structures and Algorithms in C++.
Author:	Adam Drozdek.
Publisher:	Cengage Learning, Latest Edition.

Major Topics Covered in the Course

	Topics	Hours
1	Introduction to Complexity Analysis, Big-O Notation, Recursion and Analysis of Recursive Algorithms.	6
2	Lists, Stacks & Queues.	9
3	Trees (Binary trees, Binary search trees, Tree traversal algorithms, AVL Trees, B-Trees).	9
4	Hashing (hash tables, hashing functions, collision resolution techniques).	3
5	Heaps (Max Heap, Min Heap).	3
6	Graphs (Representation, traversal algorithm, depth/breadth first search, shortest paths, minimum spanning tree).	9
7	Review & Exams.	3
	Total.	42

Grading

20%	Midterm 1
20%	Midterm 2
20%	Hmk/quiz/project
40%	Final

Course Outcome

Students should be able to:

1. Design and implement basic operations on all data structures covered in this course.
2. Implement well-known algorithms related to the course, including: common $O(N^2)$ and $O(N \log N)$ sorting algorithms; Dijkstra and Floyd's shortest paths; Prims and Kruskal's MST; Transitive closures, topological sorting.
3. Design and implement appropriate hashing functions and collision-resolution algorithms.
4. Analyze and compare time- and space-complexity of basic operations for all the data structures and algorithms covered in this course.
5. Select appropriate data structure and algorithms for solving problems.

Relationship between Course Outcomes and Student Outcomes

Course Outcomes	Student Outcomes					
	1	2	3	4	5	6
1		H				H
2		H				M
3		H				M
4	H					M
5	H					H